

ATHARV JOSHI

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Professional Summary

Associate Software Engineer with 1.5+ years delivering production GenAI systems using LLMs, Retrieval-Augmented Generation (RAG), and Autonomous AI Agents. Skilled in LangChain, LangGraph, CrewAI, FastAPI, Prompt Engineering, and vector database integration. Achieved 94% document validation accuracy, 45% reduction in manual processing, and 99.5% API uptime. Experienced with OpenAI GPT-4, Google Gemini, and Azure AI on AWS and Azure cloud infrastructure.

Key Highlights

- 45% reduction in manual processing time by architecting GenAI-powered automation pipelines.
- 94% accuracy in AI-driven packaging document validation across 500+ daily documents.
- 99.5% API uptime on production FastAPI services handling high-volume document processing.
- 42% gain in job-match search accuracy via RAG pipeline with LangChain and AWS OpenSearch.
- 35% decrease in operational overhead by implementing Autonomous AI Agents using CrewAI and LangChain.
- Mentored AI/ML interns and delivered multiple Proof of Concepts (POCs) across client engagements.

Education

Sanjivani College of Engineering, Kopergaon

Bachelor of Technology (B.Tech) — Computer Engineering

2020 – 2024

CGPA: 9.05 / 10.0

Work Experience

Associate Software Engineer — Generative AI Department

Nov 2024 – Present

Benchmark IT Solutions

Pune, India

- Architected end-to-end **GenAI** solutions using **Python** and **LLMs**, cutting manual processing time by 45% across core business workflows.
- Engineered **RAG** pipelines for end-to-end document validation using **Qdrant** and **FAISS** as vector stores, achieving 92% accuracy and reducing errors by 60%.
- Built and deployed production-grade **RESTful APIs** with **FastAPI** and **Flask**, handling 500+ daily requests at 99.5% uptime.
- Orchestrated **Autonomous AI Agents** and **multi-agent workflows** using **LangChain**, **LangGraph**, and **CrewAI**, scaling operational capacity by 35%.
- Scheduled asynchronous background tasks via **Celery** with **Redis** as message broker, ensuring zero workflow bottlenecks at scale.
- Deployed **OpenAI GPT-4** and **Google Gemini** APIs for real-time AI features, enhancing system response times by 50%.
- Mentored AI/ML interns and partnered with cross-functional teams to prototype and ship **AI-powered POCs** across multiple client engagements.

AI Engineer Intern

Sept 2023 – Jun 2024

HESA-ONE LLP

Nashik, India

- Constructed an industry-focused **CycleGAN** generative model that synthesized 5,000+ realistic images with an 85% realism score.
- Streamlined the full **ML pipeline** — data collection, preprocessing, model training, fine-tuning, and evaluation — scaling performance by 22% and reducing training time by 30%.

Projects

AI-Powered Packaging Document Validation System

Python, OpenCV, VLM, Azure AI, Azure Custom Vision, Qdrant, Redis, Celery, SMTP, Langfuse, FastAPI, Docker

- Delivered an **AI validation system** processing 500+ packaging documents daily at 94% accuracy, ensuring textual and visual compliance with approved design standards.
- Leveraged **Vision Language Models (VLMs)** and **Azure Custom Vision** to verify logo placement, icon alignment, and visual element positioning — cutting manual review by 70%.
- Mechanized validation workflows via **OpenCV** and **Azure AI**, scaling operational efficiency by 55% and eliminating 80% of manual errors.

- Integrated **Celery** task queues with **Redis** broker to process validation jobs asynchronously, ensuring non-blocking pipeline execution at scale.
- Configured **SMTP**-based programmatic email notifications to dispatch validated results and documents to the respective department head upon completion, reducing reporting delays by 65%.
- Tracked LLM pipeline quality using **Langfuse** for observability, tracing, and real-time performance monitoring in production.
- Containerized the full application using **Docker** and deployed via **CI/CD** pipelines on **GitHub Actions** for reliable production delivery.

Asgard World — AI-Powered Job Portal

LangChain, OpenAI GPT-4, Gemini, Temporal, AWS Lambda, AWS OpenSearch, FastAPI, PostgreSQL, Qdrant, RAG, Docker

- Built an **AI-driven job portal** with **CV parsing**, cover letter generation, and personalized job matching for 1,000+ candidates at an 85% relevance score.
- Deployed a **RAG pipeline** with **LangChain** and **AWS OpenSearch** for semantic profile-to-job matching, advancing search accuracy by 42%.
- Integrated **Temporal** for durable workflow orchestration, managing long-running AI tasks including CV analysis and job recommendations with guaranteed execution and fault tolerance.
- Incorporated **OpenAI** real-time API for **AI-driven interview simulations**, substantially enhancing candidate preparation experience.
- Designed a scalable **FastAPI** backend on **AWS Lambda** with **Docker**, establishing a cost-efficient serverless architecture for high-traffic scenarios.

Facial Age Progression using CycleGAN

Python, TensorFlow, PyTorch, CNN, CycleGAN, Data Augmentation

- Designed a **deep learning** model for facial aging simulation achieving 78% accuracy, preserving facial characteristics with minimal distortion.
- Applied **data augmentation** and preprocessing pipelines, boosting model accuracy by 18% and reducing overfitting by 25%.

Technical Skills

Hard Skills

Programming Languages: Python, C++, C

Generative AI and LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Generative AI (GenAI), Prompt Engineering, Function Calling, Few-Shot Prompting, Chain-of-Thought Reasoning, Embeddings, Semantic Search, Model Context Protocol (MCP)

AI Agents and Orchestration: Autonomous AI Agents, Multi-Agent Systems, LangChain, CrewAI, Temporal

Machine Learning and Deep Learning: Machine Learning (ML), Deep Learning, Natural Language Processing (NLP), Computer Vision, Generative Adversarial Networks (GANs), Supervised Learning, Unsupervised Learning

Frameworks and Libraries: FastAPI, Flask, TensorFlow, PyTorch, OpenCV, Scikit-learn, Gradio, Streamlit

Vector Databases: Qdrant, FAISS, Pinecone, ChromaDB, AWS OpenSearch

AI APIs and Platforms: OpenAI GPT-4, Google Gemini, Azure OpenAI Service, Anthropic Claude, Vertex AI

Async and Task Queue: Celery, Redis (Message Broker), SMTP (Email Automation)

Cloud and DevOps: AWS (Lambda, OpenSearch, S3), Azure AI, Azure Custom Vision

Databases: MySQL, PostgreSQL, MongoDB, Redis

Monitoring and Observability: Langfuse, LLM Tracing, AI Pipeline Monitoring

Developer Tools: Git, GitHub, JIRA, Cursor AI, Windsurf, VS Code, Claude Code

Soft Skills

Cross-Functional Collaboration | Technical Mentorship | Analytical Problem Solving | Attention to Detail | Adaptability | Clear Communication | Time Management | Self-Directed Learning | Stakeholder Engagement | Agile Teamwork